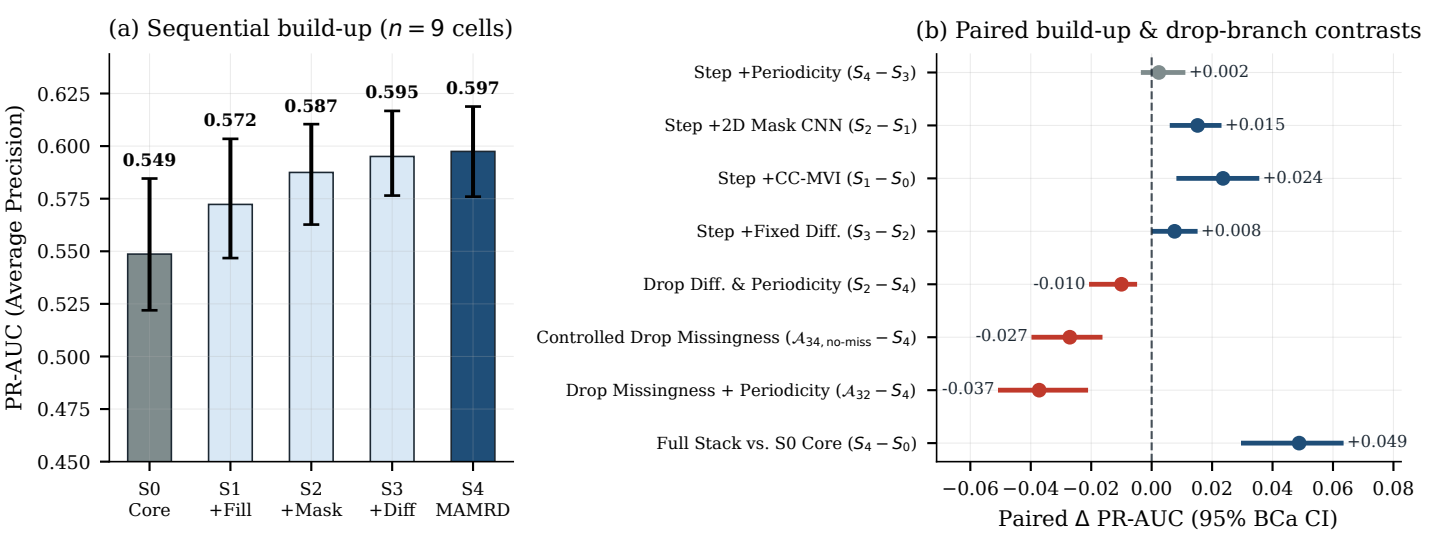
(a) Sequential build-up ($n = 9$ cells)
PR-AUC (Average Precision)
0.625
0.600
0.575
0.550
0.525
0.500
0.475
0.450
0.549
0.572
0.587
0.595
0.597
S0 Core
S1 +Fill
S2 +Mask
S3 +Diff
S4 MAMRD
(b) Paired build-up & drop-branch contrasts
Step +Periodicity ($S_4 - S_3$)
Step +2D Mask CNN ($S_2 - S_1$)
Step +CC-MVI ($S_1 - S_0$)
Step +Fixed Diff. ($S_3 - S_2$)
Drop Diff. & Periodicity ($S_2 - S_4$)
Controlled Drop Missingness ($\mathcal{A}_{34,\,\text{no-miss}} - S_4$)
Drop Missingness + Periodicity ($\mathcal{A}_{32} - S_4$)
Full Stack vs. S0 Core ($S_4 - S_0$)
+0.002
+0.015
+0.024
+0.008
-0.010
-0.027
-0.037
+0.049
−0.06 −0.04 −0.02 0.00 0.02 0.04 0.06 0.08
Paired Δ PR-AUC (95% BCa CI)